Answer key. Answers are in magenta.

NRES 746 – Week 1, Lecture 2: Pseudocoding a Permutation Test

Instructions: Work alone for about 5 minutes on Parts 1-2. Then get together with your neighbor(s) and spend about 5 minutes on Part 3, translating your pseudocode into actual R syntax **by hand, on paper** (laptops and phones put away).

Background: what is a permutation test?

In previous stats classes, you've seen tests for whether a sample's mean differed from a known null distribution: e.g., a z-test, and a t-test. These tests rely on knowing (or assuming) the shape of the population distribution the data came from.

A **permutation test** drops that requirement. It's a *nonparametric* (distribution-free) test: instead of simulating data from an assumed distribution, we generate a null distribution for our test statistic by **randomly reshuffling the labels** (e.g., treatment vs. control) on the data we already have, over and over. If the group labels really don't matter, shuffling them shouldn't change/reduce the expected difference between group means very much. If the observed difference is much larger than what typical reshufflings produce, that's evidence the labels matter.

Part 1: The scenario

Here's the scenario: pygmy short-horned lizards were sampled from a cheatgrass-removal treatment area (trtA) and a nearby untreated control area (Control). We want to know whether body mass differs between the two groups.

```
trtA:      175  168  168  190  156  181  182  175  174  179
Control:   185  169  173  173  188  186  175  174  179  180
```

1a. What is the observed test statistic (the data summary we'll compare against the null distribution)?

The difference in mean body mass: $\bar{x}_{\text{trtA}} - \bar{x}_{\text{Control}} = 174.8 - 178.2 = -3.4$ (or its absolute value, 3.4).

1b. Is the alternative hypothesis one-tailed or two-tailed ? Briefly explain why.

Two-tailed: the question is whether mass *differs*, with no direction predicted in advance.

Part 2: Write the pseudocode

2. In words, describe the algorithm you would need to run a permutation test on this data. Think about: what gets shuffled, what gets recomputed each time, how many times you repeat it, and how you turn the result into a p-value.

Compute the observed difference in means. Then, many times (e.g. 10,000): shuffle the group labels at random across all 20 lizards, recompute the difference in means for the shuffled groups, and store it. The p-value is the fraction of stored differences at least as extreme as the observed one (in absolute value, for a two-tailed test).

Part 3: Translate to R (group, pen and paper)

3. As a group, fill in the blanks below to turn your pseudocode into real R syntax. You don't need working code – just your best attempt at the correct functions and syntax, written by hand.

```
permutation_test <- function(data, group, value, reps){
```

```
  observed_dif <- ____________________________________________

  null_difs <- ____________________________________________

  for(i in 1:reps){

    shuffled_labels <- ____________________________________________

    null_difs[i] <- ____________________________________________

  }

  p_value <- ____________________________________________

  return(p_value)
}
```

Flag any particular points of confusion that you'd like to come back to for discussion!

```
permutation_test <- function(data, group, value, reps){
  labs <- unique(data[[group]])
  observed_dif <- mean(data[[value]][data[[group]] == labs[1]]) -
                  mean(data[[value]][data[[group]] == labs[2]])
  null_difs <- numeric(reps)
  for(i in 1:reps){
    shuffled_labels <- sample(data[[group]])
    null_difs[i] <- mean(data[[value]][shuffled_labels == labs[1]]) -
                    mean(data[[value]][shuffled_labels == labs[2]])
  }
  p_value <- mean(abs(null_difs) >= abs(observed_dif))
  return(p_value)
}
```